

Team of Teams & the Training of Staffs and Operational Planners

What IT portfolio and solution levels can learn from the military

Version 4.0 · As of 2026-08-13 · Audience: decision-makers without detailed knowledge of the subject – no military or agile background required · Companion document to the situational-awareness reading list

About this document: Curated, conceived and editorially owned by **Robert Seebauer**; researched and produced with the support of **Claude** (Anthropic – model **Claude Fable 5**, via Claude Code). This is **version 4.0** (addition: Annie Duke, *Thinking in Bets*, as the fifth additional core source in Part A2.5 – the anti-resulting rule for the AAR culture, forecast literacy, calibration by betting and Ulysses contracts in the curriculum; sister documents: EA memorandum v2.0 and AI memorandum v1.1). Version management and previous editions: Quellen/Versionierung.md and Quellen/Versionen/.

Executive Summary

What is this about? For more than 200 years, military organizations have systematically trained **staffs** (teams that prepare leadership decisions) and **operational planners** (specialists in planning complex undertakings across many units). Corporations steering IT portfolios across many teams, departments and suppliers face the structurally identical problem: **many semi-autonomous units must be aligned on a common goal – under uncertainty, with incomplete information and a constantly changing situation.**

1

From “Team of Teams”

Scaling succeeds not through more control but through a **shared operating picture** (“shared consciousness”) plus **decentralized decision authority** (“empowered execution”). Both together – either one alone is ineffective or even dangerous. The key instrument: a daily, radically open situation briefing – the exact counterpart of our Situation Report.

2

From staff training

Militaries train planners primarily not in **tools** but in **methods of thinking**: structured decision processes, red teaming against one’s own assumptions, a common language and doctrine. The training is long (1–2 years full-time for top planners), exercise-based and selective.

3

For our portfolio

The transferable building blocks are concrete: a shared situation-picture ritual, a standardized planning process for large undertakings, liaison roles, After Action Reviews as a learning routine – and an internal training curriculum for “portfolio planners” (Part D: a concrete 12-month proposal).

THE SINGLE MOST IMPORTANT SENTENCE IN THIS DOCUMENT

It is not the reporting tool that creates the operating picture, but the **ritual** and the **trained people** around it – the tool (our software) merely makes the ritual cheaper, faster and more honest.

Part A – The Source: “Team of Teams” (McChrystal, 2015)

A.1 The book and its starting problem

Stanley McChrystal, Tatum Collins, David Silverman, Chris Fussell – *Team of Teams: New Rules of Engagement for a Complex World* (Portfolio/Penguin, 2015; German edition by Vahlen). General (ret.) McChrystal led the Joint Special Operations Command (JSOC) in Iraq 2003–2008 – a network of elite units, intelligence services and partner agencies.

JSOC was the best-trained, best-equipped and best-funded military organization in the world – and was still losing against a poorly equipped but networked, fast-learning adversary network. McChrystal’s diagnosis: the problem was not competence but **organizational architecture** – an efficient but slow hierarchy of excellent individual teams that did not know each other, did not share information, and waited for approvals from above. His famous shorthand: *“It takes a network to defeat a network.”*

This diagnosis translates almost verbatim to corporate IT: excellent individual teams (development, operations, security, business units) optimizing within silos, while the overall undertaking – the *large solution* – appears slow and uncoordinated.

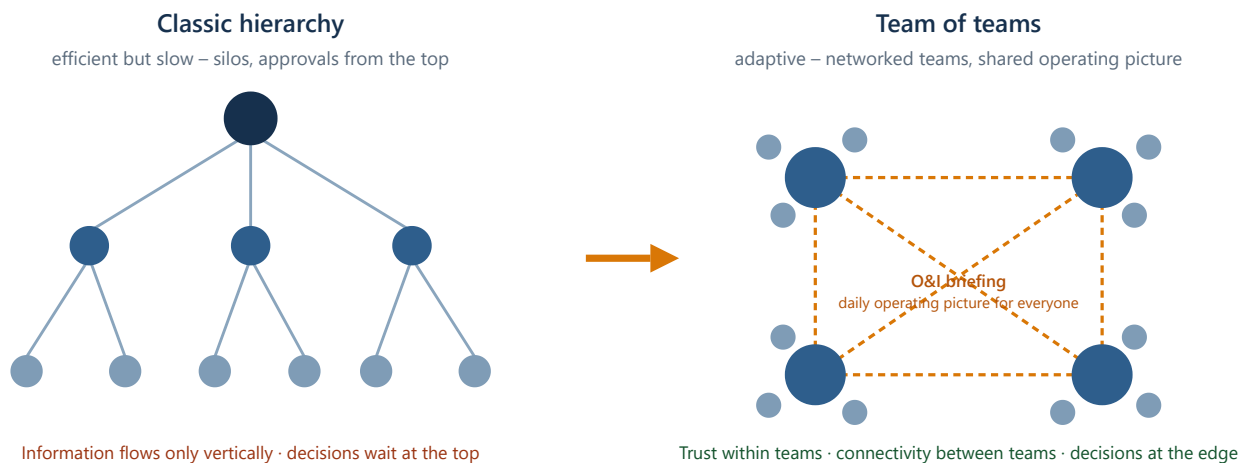


Fig. 1 – From command hierarchy to the networked “team of teams”: existing teams remain, but are connected through rituals (O&I) and liaison roles.

A.2 The four core concepts

1 Complicated is not complex

Complicated systems (a clockwork, an assembly line) have many parts but are predictable – classic management works here: plan, optimize, control. Complex systems (weather, markets, software portfolios with dozens of teams) have *interacting* parts – small causes produce large, unpredictable effects. The consequence: in complex environments, **adaptability beats efficiency** and **resilience beats prediction**. Anyone who steers a complex portfolio like a complicated project (detailed plan, milestone control) produces an illusion of certainty.

2 Shared consciousness – the shared operating picture

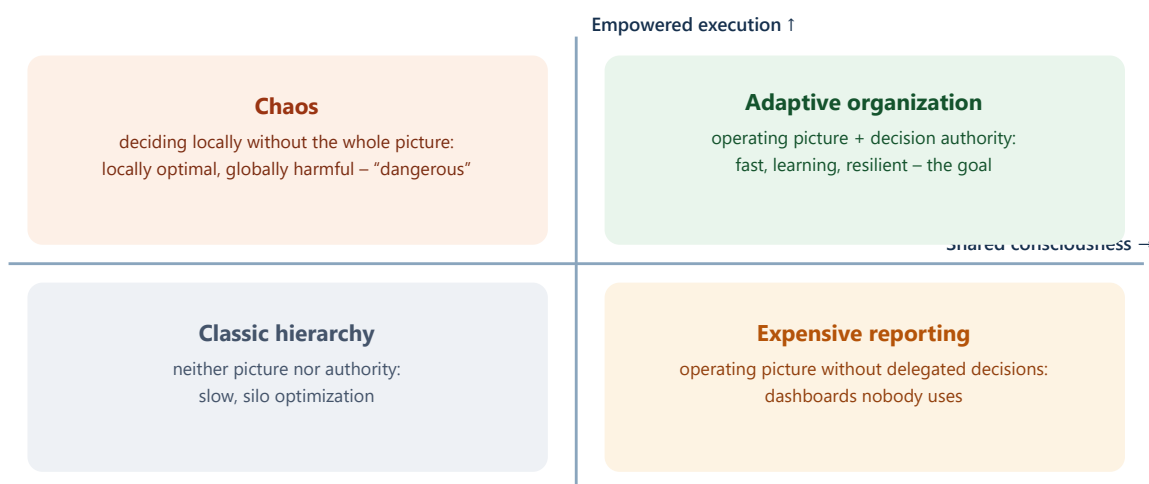
Everyone in the organization must understand the *whole picture*, not just their own slice. JSOC's central instrument: the daily **O&I briefing** (Operations & Intelligence) – a video-broadcast situation briefing, ~90 minutes, at times with **several thousand participants** across all levels and partner agencies. What mattered: **radical transparency instead of need-to-know** (information is shared by default), **leadership demonstrating the culture** (McChrystal moderated himself, asked open questions, thanked people for bad news), and **rhythm beats perfection** (daily and current matters more than complete and polished).

3 Empowered execution – decentralized decisions

Decisions are moved to where the best information sits – down and forward. The leadership principle: **“eyes on – hands off”**: leadership sees everything (through the shared picture) but does not intervene in every decision. The book's most important warning: **empowerment without a shared operating picture is dangerous** – people who are supposed to decide locally without knowing the whole picture produce locally optimal, globally harmful decisions. Conversely, an operating picture without decision authority is just expensive reporting.

4 Connection architecture instead of reorganization

JSOC was *not* merged into one giant team (trust and shared purpose only work in small groups). Instead, existing teams were deliberately networked: **liaison officers** – deliberately the *best* people (not the most expendable!) were sent to partner units for six months; an **embedding program** had team members work in other teams for weeks at a time; top leadership shifted from **chess master to gardener**, tending the ecosystem: rituals, information flows, culture.



The McChrystal coupling: only the top-right quadrant works – both together or not at all.

Fig. 2 – Why one element alone is not enough: shared picture and decentralized decisions depend on each other.

A.3 Direct relevance for our solution/portfolio feature

Team-of-Teams concept	Counterpart in our context
O&I briefing	Aggregated Situation Report + a regular portfolio/solution sync across all ARTs/teams
Shared consciousness	One consolidated operating picture instead of many silo reports; drill-down accessible to everyone
Empowered execution	Guardrails + decentralized prioritization instead of central case-by-case approvals
Liaison officers	Named interface roles between ARTs/value streams (not just tickets!)
Eyes on – hands off	Leadership reads the picture, escalates only on guardrail violations
Complicated ≠ complex	Monte Carlo forecasts (our Simulate module) instead of single-point predictions

Part A2 – Four Additional Core Sources (new in v2.0)

The following five works each deepen one aspect of the Team-of-Teams findings: **Clausewitz** provides the theory of why plans grind against reality; **Marquet** the concrete operating system for decentralized leadership; **Pink** the motivational-psychology explanation of why it works; **Hubbard** the measurement craft for an honest operating picture; **Duke** (new in v4.0) the hygiene of judgment with which decisions and outcomes are assessed separately.

A2.1 Carl von Clausewitz – *On War* (posthumous, 1832)

THE foundational work on deciding and acting under uncertainty – required reading in practically every general-staff education to this day, and (via Moltke) the intellectual root of mission command. The load-bearing concepts and their transfer:

- **Primacy of policy:** war is the continuation of policy by other means → the portfolio is the continuation of corporate strategy; every initiative must be able to name its strategic purpose.
- **Friction:** countless small, individually trivial resistances add up; the plan grinds against reality → the gap between roadmap and reality is a property of the system, not a management failure; the answer is buffers, short feedback cycles and adaptability – not "better" detailed plans.
- **Fog of uncertainty:** much decision-relevant information is uncertain or missing → no operating picture is complete; declare confidence and data quality instead of false precision.
- **Purpose – goal – means:** a clean hierarchy of why/what/with what → the intent cascade from strategy to team mission; OKRs are a modern form of it.
- **Center of gravity (Schwerpunkt):** concentrate forces at the decisive point → real prioritization and WIP limits; a portfolio doing "everything at once" has no Schwerpunkt.
- **Culminating point:** every offensive loses force as it extends → initiatives and scale-ups overextend (teams, integration effort, quality); early indicators of this belong in the operating picture.
- **Moral factors:** will, morale and trust are real factors of strength → team health/DevEx are steering variables of equal rank with flow metrics.

A2.2 L. David Marquet – *Turn the Ship Around!* (2012)

Marquet took command of the USS Santa Fe – then the worst-rated submarine in the US Pacific Fleet – and made it the best-rated within a year by inverting the leadership model: **leader-leader instead of leader-follower**. Core idea: do not move information to authority; **move authority to the information**. Three levers:

- **Give control:** shift decision authority downward step by step; the most visible ritual is the language ladder – subordinates say "**I intend to ...**" instead of "What should I do?" → teams report intentions in the portfolio sync instead of filing approval requests; the board intervenes only on guardrail conflicts.
- **Build competence:** delegation without competence is negligent; continuous technical training and deliberate action → exactly the gap our curriculum (Part D.2) closes.
- **Create clarity:** everyone must know the organization's purpose and priorities, or decentralized decisions diverge → commander's intent, strategic themes, a shared operating picture.

The operational practice book for mission command in a civilian context – the most concrete guide to actually introducing "empowered execution".

A2.3 Daniel H. Pink – *Drive* (2009)

Pink synthesizes decades of motivation research (Deci/Ryan et al.): the classic incentive model of reward and sanction ("Motivation 2.0") works for algorithmic routine work – for heuristic knowledge work, if-then rewards demonstrably narrow focus, crowd out intrinsic motivation and worsen performance. What works is "Motivation 3.0": **autonomy** (self-direction over task, time, technique, team), **mastery** (getting better at something that matters) and **purpose** (contributing to something larger than oneself).

Transfer: the motivational-psychology **explanation** of why mission command, empowered execution and leader-leader work – and a warning for the metrics system: tie velocity or metric targets to bonuses and you harvest Goodhart effects instead of performance. Autonomy ↔ decentralized decisions, mastery ↔ planner training, purpose ↔ intent.

A2.4 Douglas W. Hubbard – *How to Measure Anything* (2007/2014)

Core thesis: **anything that influences a decision is measurable** – measurement does not mean perfection but **reduction of uncertainty**. Tools:

- **Calibrated estimators**: with a few hours of training, experts learn to state 90% confidence intervals that are actually right 90% of the time – estimating becomes a testable skill.
- **Fermi decomposition**: break large unknowns into estimable parts.
- **Rule of Five**: just 5 random samples give a >93% probability interval around a population median – small measurements beat no measurement.
- **Expected Value of Information (EVI)**: measure only what can change decisions – the sharpest antidote to KPI graveyards.
- **Applied Information Economics**: Hubbard's overall method – model the decision, estimate calibrated, compute the value of information, measure selectively, decide via Monte Carlo.

Transfer: the measurement foundation of our operating picture – confidence intervals instead of point forecasts (exactly the principle of the Simulate module), EVI as the selection criterion for report metrics, calibration training as a building block of curriculum module 3.

A2.5 Annie Duke – *Thinking in Bets* (2018) (new in v4.0)

The former world-class poker player with a cognitive-science background supplies the fifth building block: the **hygiene of judgment** under uncertainty – for what can still go wrong after operating picture, plan and forecast is the judging itself.

- **Life is poker, not chess:** real decisions are made on incomplete information and under the influence of luck; whoever grades them like chess moves (full transparency, no chance) draws the wrong lessons – Clausewitz's fog from the player's perspective.
- **Resulting:** the most widespread thinking trap – measuring decision quality by outcome quality. A good decision plus bad luck is punished as a mistake, a bad decision plus good luck is celebrated as a success; the organization learns randomness instead of skill.
- **"Wanna bet?":** anyone who would have to bet on a conviction automatically states probabilities instead of certainties – "being wrong" becomes an update of degrees of belief instead of a loss of face. The everyday form of Hubbard's calibration (A2.4).
- **Truthseeking groups:** decision quality needs groups with an explicit truthseeking charter – sharing information, organized skepticism, accountability, diversity of opinion. The staff as a truthseeking group is the social form of red teaming.
- **Mental time travel:** backcasting and premortem as well as **Ulysses contracts** – advance commitments that bind the future self before pressure and emotion decide.

Transfer: (1) the **anti-resulting rule** as the second culture rule of AAR practice (B.3, competency field 5, module 6). (2) **Forecast literacy** (competency field 1): A 15% scenario that materializes does not refute a P85 forecast. (3) The **betting ritual** as everyday calibration in module 3. (4) Decision points with a pre-agreed response are **Ulysses contracts** – the decision-point wake-up call of the AI memorandum (M4 there) is their technical implementation.

How it fits: Duke complements Hubbard – he makes estimating testable, she makes judging testable.

Part B – How Militaries Train Staffs and Operational Planners

B.1 What is a staff – and why does it matter?

A **staff** is the team that prepares the decision basis for a leader (commander): reconnoiter and condense the situation, develop options, think through consequences, formulate orders, monitor execution. The commander decides – the staff makes sure the commander decides *well-informed* and that the decision is communicated in an *executable* way.

The corporate analogy is immediate: a portfolio board, a PMO, a solution-management team **is** a staff. The difference: militaries treat staff work as a distinct, teachable **craft with multi-year training** – corporations often fill the same functions with subject-matter experts who have never received any methodical planner training, and then wonder about meeting chaos, slide battles and decision gridlock.

The historical core: the Prussian-German **General Staff** (from Scharnhorst, ~1810) was the first institution to systematically replace “brilliant improvisation” with **trained method** – with the War Academy as its school, a unified doctrine, and the principle of **Auftragstaktik** (internationally “mission command”): the leader specifies *goal and intent* (“what and why”); the level below decides the “how” itself. Mission command is the historical original of McChrystal’s “empowered execution” – and of goal steering via OKRs.

B.2 The most important training institutions

Bundeswehr Command and Staff College (Führungsakademie, Hamburg) – LGAN DE · 2 years

The German general/admiral staff course (LGAN) is Germany’s top-tier training: 2 years full-time, ~100 national and international participants per year; only roughly the best ~18% of an officer cohort get in. Content: leadership and ethics, operational planning at all levels, security policy, cooperation with ministries and allies. Essential: participants learn to **work in mixed staffs** – not merely to accumulate individual knowledge. (*Deepened in v2.0*) The LGAN is only the top of a **tiered system**: the **basic staff officer course (BLS)** qualifies *every* future staff officer in the staff craft; the two-week **refresher course (SFL)** follows two years later; the **LGAI** (12 months, nicknamed the “little UN”) has trained over 3,000 officers from more than 100 nations – international networking is an explicit training goal. *Lesson: broad training for everyone + elite training for a few + regular refreshers, not just a single elite program.*

United Kingdom – Defence Academy / JSCSC (Shrivenham): ACSC UK · ~46 weeks · new in v2.0

The Joint Services Command and Staff College consolidates staff training of all three services – deliberately **joint**. Its core product is the **Advanced Command and Staff Course (ACSC)**: ~46 weeks (42 weeks of training plus block leave), annually from September, for officers at the top of their cohort, with participants from **over 50 nations**. The declared aim is developing “effective intellect”: analytical decision-making **plus** communication skills, broadened by strategy, security studies and multi-domain planning; academic support traditionally comes from King’s College London, and many graduates earn a master’s degree. Beneath it sits a tiered pipeline (Intermediate Command and Staff Course for the broad base, Higher Command and Staff Course for future general officers). *Lesson: cross-functional mixed cohorts (train IT + business + finance together), academic partnership, a three-tier pipeline instead of a single course.*

Austria – National Defence Academy Vienna: General Staff Course AT · new in v2.0

Austria's general-staff training is remarkable for its **selection and probation design**: about a year before the course starts, preparatory courses lead to a **first examination** (tactics, military fundamentals, general knowledge – only ~20–25% pass) and a subsequent **main examination** at the academy (~50% pass rate) – **with no retake**. Then comes the multi-year course (in today's structure two years plus a **probation year in general staff service**; earlier cohorts ran three years). Content has been modernized around cyber security, information warfare and international cooperation; guest participants come from Switzerland, Germany and France, among others. *Lesson: hard, fair selection BEFORE the expensive training – and a real probation phase in the target role AFTER it, before the status is finally conferred.*

US Army Command and General Staff College (CGSC, Fort Leavenworth) US · ~10 months

The broad training for staff officers. Its centerpiece is the **MDMP** – the “Military Decision-Making Process”, a standardized 7-step planning process (B.3). More important than the specific process is the principle: **all staff officers of all units learn the same process and the same language** – any newly assembled staff is thereby immediately able to work.

School of Advanced Military Studies (SAMS, Fort Leavenworth) US · 11 months · elite

The elite school *after* CGSC: a graduate-level program that turns selected officers into professional **operational planners** (nicknamed “Jedi Knights”). Its educational goal reads like a job description for portfolio strategists: critical and creative thinkers who help senior leaders **understand the operational environment** and **visualize and describe solutions to complex problems**. The curriculum integrates military history, theory and doctrine with practice – roughly half the time is planning exercises. Graduates serve a mandatory tour as leads of **operational planning teams** – training is directly coupled to a role.

University of Foreign Military and Cultural Studies (“Red Team University”) US · 2004–2021

Trained **red teamers**: specialists who attack assumptions, force alternative perspectives and prevent groupthink. Core of the curriculum: applied critical thinking, cultural empathy, self-reflection, groupthink mitigation via anonymous feedback and structured facilitation formats. Underlying premise: **organizations court failure in predictable ways, gradually, along their own thought patterns and culture**. The freely available *Red Team Handbook / Applied Critical Thinking Handbook* documents ~50 techniques (premortem, “Four Ways of Seeing”, assumption checks, devil's advocate with a mandate) – the single most directly transferable source in this briefing. (Defunded in 2021 for budget reasons; the methodology lives on in civilian programs, see Part D.)

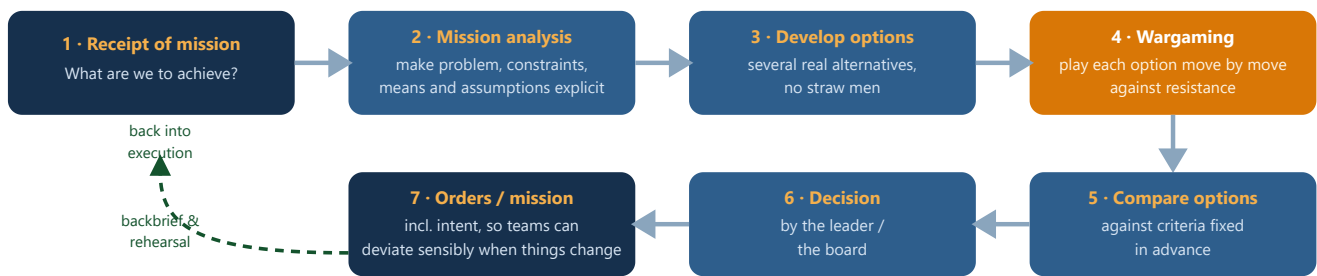
NATO – Comprehensive Operations Planning Directive (COPD) NATO

Standardizes operational planning across 30+ nations – proof that a common planning language works even across organizational *and* national borders, provided it is trained and exercised.

WHAT ALL SYSTEMS EXAMINED (DE, US, UK, AT) HAVE IN COMMON – NEW IN V2.0

(1) a **tiered pipeline** – broad training for all staff roles, elite training for a few; (2) **selection with consequences** – examinations that can actually be failed; (3) **mixed cohorts** – joint and international, because networking is itself a training goal; (4) **1–2 years of seriousness** instead of one-day seminars; (5) **role coupling** – training is mandatorily followed by a planner assignment, in Austria even a formal probation year. Each of these five elements transfers to a corporate curriculum (Part D.2).

B.3 What exactly do operational planners learn? The methods in plain language



The value lies not in the 7 steps themselves, but in everyone mastering the same process and the same language.

Fig. 3 – The Military Decision-Making Process (MDMP) as a reference model for structured portfolio planning.

1. The structured planning process (MDMP as reference model)

Seven steps (Fig. 3): receipt of mission → **mission analysis** (What actually is the problem? Mission, constraints, means, assumptions) → **develop options** (several real “courses of action”, not one preferred solution plus two straw men) → **play the options through** (wargaming; this produces synchronization matrices and decision points) → **compare options** (against explicit, pre-agreed criteria) → decision → **formulate the order** – including the commander’s intent, so that subordinates can deviate sensibly when the situation changes.

2. Wargaming / planning exercises

The central technique for both training *and* planning: a plan is played through against an actively thinking adversary/resistance (“action – reaction – counteraction”). Purpose: find weaknesses *before* reality finds them; identify decision points and early indicators: “How do we recognize early that plan A is tipping – and what do we do then?”

3. Red teaming / critical thinking

Institutionalized dissent with a mandate: make assumptions explicit and attack them one by one; the **premortem** (“It is 12 months later and the undertaking has failed – why?”); think the situation through from the other actors’ perspective; collect anonymous assessments *before* group discussions so the highest-ranking person does not shape the outcome.

4. After Action Review (AAR)

After every exercise and every operation, guided by four questions: **What was planned? What actually happened? Why the difference? What exactly do we change?** The culture is decisive: **rank-free** (the private may contradict the colonel), timely, documented – and the findings flow into a central clearing house (US Army: Center for Army Lessons Learned) so that the *organization* learns, not just the individuals involved. *Addition in v4.0 (Duke)*: the second culture rule is the **anti-resulting rule** – decision quality and outcome quality are assessed separately: A good outcome does not ennoble a bad decision, a bad one does not condemn a good one; otherwise the organization learns good and bad luck instead of skill.

5. Backbrief and rehearsal

After orders are issued, the executing level explains back *in its own words* how it understood mission and intent (backbrief) – misunderstandings surface immediately. Before execution, the sequence is walked through together (rehearsal / “ROC drill”).

6. Common language, doctrine, formats

Standardized terms, symbols, report formats and outlines – not out of love for bureaucracy, but so that under time pressure nobody has to think about *formats* and every report is instantly readable.

The didactics behind it: consistently **exercise-based** (planning exercises, command post exercises, simulation centers), **historically grounded** (case studies, “staff rides”), **selective** (only a fraction become planners), **role-coupled** (training is mandatorily followed by a planner assignment) and **career-relevant** (general staff training is a promotion prerequisite – the organization signals: planning is top-tier work, not a siding).

Part C – Transfer: What Must People in IT Portfolios Learn?

C.1 Role mapping – who is “the staff” in our world?

Military role	Corporate/portfolio counterpart
Commander	Portfolio owner / division executive / LPM board
Chief of staff	Head of PMO / head of portfolio management
Operational planner (G5/J5)	Portfolio/solution managers, solution train engineer, lead architects
Situation officer (G2/G3)	Owners of situation reports, metrics, forecasts
Liaison officer	Named interface role between ARTs / value streams / business-IT
Liaison officer to the superior staff (<i>new in v3.0</i>)	"Architect Elevator" (Hohpe): lead/solution architects as the vertical liaison between the board ("penthouse") and engineering ("engine room")
Red team	Independent challenger role on the portfolio board (can rotate)

C.2 The competence catalogue – what participants must learn

1 Creating and reading the operating picture (situational awareness)

Condense data into an honest picture; understand flow metrics, forecasts and confidence; know what a metric does *not* say; declare data quality instead of false precision; read probabilistic statements without “resulting” (Duke): A 15% scenario that materializes does not refute a P85 forecast.

Typical deficit: status reports as justification – “watermelon reports”: green outside, red inside.

2 Structured planning (an “MDMP for portfolios”)

For every large undertaking: clean problem analysis before solution debate; **at least two real options** with explicit comparison (cost, risk, time-to-market, reversibility); document assumptions; formulate intent so that teams can act sensibly on their own when the situation changes.

Typical deficit: business cases with exactly one option and no documented assumptions.

3 Stress-testing plans (wargaming & red teaming)

Premortems before major commitments; scenario runs (“supplier fails / key team quits / budget –20% – then what?”); define decision points and early indicators; be able to facilitate groupthink countermeasures.

Typical deficit: risks are listed (Excel graveyard) but never played through.

4 Leading decentrally (mission command)

Formulate goals and guardrails instead of dictating measures; demand backbriefs; make escalation criteria explicit (“eyes on – hands off”); tolerate teams proceeding differently than you would.

Typical deficit: steering committees deciding implementation details for which others hold the information.

5 Organized learning (AAR culture)

Facilitate AARs after every PI/quarter/undertaking; establish rank-free candor; **separate decision quality from outcome quality (avoid resulting, Duke)**; collect findings centrally and feed them back into standards.

Typical deficit: “lessons learned” slides nobody ever opens again – and reviews that assess only outcomes instead of decisions.

6 Communicating like a staff

Short, standardized situation formats (military “BLUF” – Bottom Line Up Front: key message first); storytelling with data; the ability to present situation + options + recommendation to a board in 5 minutes.

Typical deficit: 60-slide decks without a recommendation.

C.3 The five success factors – what matters most

1. **Both together or not at all:** operating picture without delegated decisions = expensive reporting. Delegation without an operating picture = chaos. The McChrystal coupling is the core (Fig. 2); introducing only one gets you neither.
2. **Rituals before tools:** the O&I briefing was video + discipline + demonstrated culture. Our software delivers the consolidated picture – but without a fixed, chief-moderated ritual (e.g. a fortnightly portfolio sync with mandatory attendance) it remains a dashboard nobody looks at.
3. **Send your best people:** fill liaison and planner roles with top performers and make it career-relevant. Militaries make planner training a promotion prerequisite; corporations often send whoever happens to be free – that signal is received, in both directions.
4. **Process standard beats process perfection:** the MDMP’s value lies not in its seven steps but in *everyone* having learned the same process and language. A decent, universally mastered portfolio planning standard beats a brilliant one only one department knows.
5. **Institutionalize dissent:** criticism must not depend on individual courage. Red-team techniques (premortem, anonymous pre-votes, a named devil’s advocate with a mandate) turn dissent into a *role* instead of a career risk.

C.4 Four civilian corroborating witnesses from architecture practice (new in v3.0)

The theses of this briefing come from the military – which makes it all the more significant that civilian software-architecture practice independently arrives at the same findings. Four thought leaders provide the corroborating evidence (full source profiles and the EA transfer: Enterprise-Architektur-und-Solution-Lagebild_v2.0.md, Part A2 – here only what carries for staff work, rituals and training).

C4.1 Luke Hohmann – Prioritisation as a serious game (civilian wargaming)

Hohmann (*Innovation Games*, 2006; *Beyond Software Architecture*, 2003; *Software Profit Streams*, 2023) developed structured decision **games** for the portfolio: in **"Buy a Feature"** stakeholders receive a limited play budget and buy features together – wish lists turn into real trade-off conversations because nobody can afford everything. **"Prune the Product Tree"** structures roadmap conversations; **participatory budgeting** scales the principle to portfolio budgets (many voices → one legitimised decision) and was adopted by SAFe into Lean Portfolio Management.

What it means for this briefing: this is the civilian, immediately purchasable form of the planning exercises from B.3 – rituals that combine **shared consciousness** (everyone sees the same trade-offs on the same board) and **empowered execution** (participants co-decide instead of merely receiving decisions) in one format. Directly usable in the portfolio sync, purchasable as a program (D.1), and as an exercise format in the curriculum (module 2).

C4.2 Gregor Hohpe – the elevator: liaison in the vertical

Hohpe (*The Software Architect Elevator*, 2020; architectelevator.com) describes architects as **elevator riders between the penthouse (board) and the engine room (engineering)**. This briefing knows liaison officers between units – the *horizontal* networking; Hohpe adds the **vertical**: without people who regularly enter both worlds and translate in both directions, strategy and delivery drift apart without anyone noticing – every level takes its own slice for the whole picture (McChrystal's silo diagnosis, turned vertical). His second guiding sentence: *"Architects live in the first derivative"* – what counts is the rate of change, not the state. (His catalogue *Enterprise Integration Patterns*, 2003, is at the same time the doctrine principle applied to integration technology: 65 named patterns as a shared professional language.)

What it means for this briefing: (1) the role mapping (C.1) needs the **vertical liaison** in addition to the horizontal one – and success factor 3 applies here too: staff it with the best, make it career-relevant. (2) For competency field 1: a situation report that shows only states omits the derivative – **trends, accelerations and tipping points** belong in the picture.

C4.3 Martin Fowler – the civilian proof of "process standard beats process perfection"

Fowler (*Patterns of Enterprise Application Architecture*, 2002; martinfowler.com) achieved with named design patterns (Domain Model, Data Mapper, Unit of Work ...) the same thing the MDMP achieved in the military: a **shared, taught vocabulary** makes arbitrarily assembled teams immediately effective – nobody has to negotiate a private language first. That is success factor 4, independently proven by twenty years of industry practice. Add the **design-stamina hypothesis** (*Refactoring*, 2nd ed. 2018): neglected internal quality measurably throttles speed – an honest situational picture therefore also needs the debt dimension (architecturally deepened in the EA memorandum, situational-picture principle 2).

What it means for this briefing: whoever introduces the company's own planning standard (module 2, artifact "portfolio planning standard v1") argues not only with the military but with the pattern movement of the software industry – the same principle, two independent worlds.

C4.4 Robert C. Martin – practice culture and discipline as a prerequisite for speed

Martin (*Clean Code*, 2008; *Clean Architecture*, 2017) stands for the craftsmanship movement, which **institutionalised deliberate practice**: code **katas** and coding **dojos** – small, repeated exercise forms with feedback – are the civilian counterpart to planning exercises and command post exercises ("consistently exercise-based", B.3). His guiding sentence "*The only way to go fast is to go well*" summarises what general-staff training teaches implicitly: discipline and mastered form are not a brake but the prerequisite for speed under pressure (cf. B.3.6: formats exist so that nobody has to think about form under time pressure). (*Framing: deliberately normative – use as doctrine, not as a blueprint; cf. EA memorandum A2.3.*)

What it means for this briefing: for the curriculum this means: interleave **short, repeated planning "katas"** between the big modules – a situation analysis in 30 minutes, a premortem in 60 – until the form sits. Practice becomes routine instead of an event.

Part D – Concrete Training Programs

D.1 Externally available programs (purchasable)

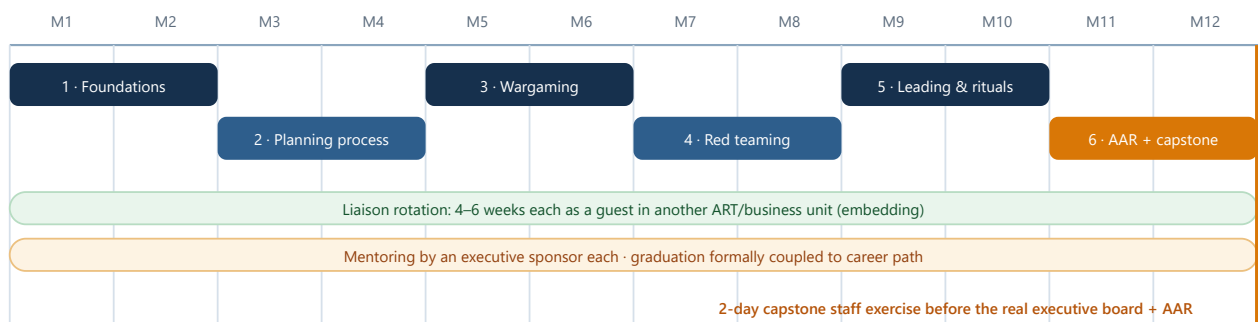
Program	Provider	Content / origin	Suitable for
Team-of-Teams consulting, red-team workshops	McChrystal Group (the author's firm)	Direct implementation of the book's concepts in corporations; builds permanent internal red-team capabilities	Executive level, portfolio boards
Red Team Thinking (training & certification)	Bryce G. Hoffman (graduate of the Army red team program)	Civilian version of the Army curriculum: assumption checks, premortem, groupthink mitigation; book <i>Red Teaming</i> (2017)	Planners, PMO, strategy
Red Team Handbook / Applied Critical Thinking Handbook	US Army (free PDF)	~50 documented techniques for self-study; free basis for internal workshops	Everyone; ideal first read
SAFe Lean Portfolio Management (3 days)	Scaled Agile	Portfolio kanban, guardrails, prioritization (WSJF), lean budgets	Portfolio managers, controllers
Flight Levels Academy	Klaus Leopold (German-language)	Operating picture & steering across three levels	Portfolio/coordination roles
Business wargaming	e.g. Gilad (<i>Business War Games</i>), specialized consultancies	Competitive/scenario simulations for strategy decisions	Strategy, M&A, major initiatives
Intent-Based Leadership	Marquet organization	Mission command for civilians: leading with intent instead of instruction (<i>Turn the Ship Around!</i>)	Line and solution leadership
Calibration training / AIE seminars (<i>new in v2.0</i>)	Hubbard Decision Research	Calibrated probability/interval estimation (a few hours, available online; ~80% of trainees reach calibration); Applied Information Economics	Everyone who estimates/forecasts: planners, architects, controllers
Superforecasting workshops (<i>new in v2.0</i>)	Good Judgment Inc. (Tetlock)	Forecasting skill built on the empirically proven levers training, teaming, aggregation; open online courses	Portfolio boards, strategy, risk
Problem Solving & Decision Making (<i>new in v2.0</i>)	Kepner-Tregoe	The classic of structured problem and decision analysis (a corporate standard for decades); the civilian counterpart to military mission analysis	PMO, operations, incident/problem mgmt
Complexity/sensemaking training (<i>new in v2.0</i>)	The Cynefin Company (Dave Snowden)	Cynefin framework: distinguish ordered from complex situations and decide accordingly – operationalizes "complicated ≠ complex"	Executives, portfolio/solution managers

Innovation Games / participatory budgeting (new in v3.0)	Applied Frameworks (Luke Hohmann; methodology of his company Conteneo)	Serious games as decision rituals: "Buy a Feature" (stakeholders buy with limited budgets), "Prune the Product Tree"; participatory budgeting for portfolio budgets – adopted by SAFe into Lean Portfolio Management (cf. C.4)	Portfolio boards, PI planning, budget rounds
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AAR in a civilian context – foundational management read: Darling/Parry/Moore, "Learning in the Thick of It", *Harvard Business Review*, July–Aug 2005.

D.2 Proposal: an internal 12-month curriculum "Portfolio Staff Training"

Goal: develop 8–12 selected people (portfolio/solution managers, PMO, lead architects, controllers) into methodically trained "portfolio planners" – analogous to SAMS: part-time alongside the job, exercise-heavy, with a mandatory assignment afterwards. **Effort:** ~15–20% of working time over 12 months; afterwards assignment as planning lead for one major undertaking each.



Part-time, ~15–20% of working time · 8–12 participants · followed by a mandatory planning-lead assignment

Fig. 4 – The 12-month "Portfolio Staff Training" curriculum at a glance.

Module 1 · Months 1–2 Foundations – complexity & the operating picture

Reading: *Team of Teams*; Dörner, *The Logic of Failure*; Meadows. **Exercise:** map your own portfolio as a complex system (dependencies, feedback loops); dissect a current status report critically: what is operating picture, what is justification?
Learning goal: participants recognize when classic project steering fails, and why.

Module 2 · Months 3–4 The planning process

Transfer MDMP principles to portfolio planning: problem analysis, real options, explicit comparison, documented assumptions, intent formulation. **Exercise:** plan a real upcoming epic to this standard – including a backbrief by the implementing team. *Addition in v3.0:* the option comparison is additionally run as a "**Buy a Feature**" session (Hohmann, cf. C.4) with real stakeholders – limited play budgets force real trade-off conversations instead of wish lists.

Artifact: the company's own one-page planning format ("portfolio planning standard v1")

Module 3 · Months 5–6 Wargaming & scenarios

Techniques: action–reaction–counteraction, decision points, early indicators; Monte Carlo forecasts as quantitative wargaming (directly executable with our Simulate module). *Addition in v2.0:* the module is preceded by **calibration training** (Hubbard, a few hours, cf. D.1) so participants' estimates and confidence intervals are dependable. *Addition in v4.0 (Duke):* beliefs are consistently phrased as **bets** ("How much would you stake on that?") – the everyday form of calibration; decision points with a pre-agreed response are documented as **Ulysses contracts**. **Exercise:** play the epic planned in module 2 through a half-day wargame against 3 disruption scenarios.

Learning goal: "How do we recognize early that the plan is tipping – and what do we do then?" becomes a standard question.

Module 4 · Months 7–8 Red teaming

Working basis: the Red Team Handbook (free); optionally an external trainer (D.1). **Techniques:** premortem, assumption check, Four Ways of Seeing, anonymous pre-votes. **Exercise:** red-team review of a real upcoming portfolio decision; participants facilitate themselves.

Artifact: the premortem becomes a mandatory step before every portfolio commitment above a defined threshold

Module 5 · Months 9–10 Leading with intent & operating-picture rituals

Mission command / intent-based leadership (Marquet); design of your own "O&I": rhythm, participants, facilitation rules, role of the software (the Situation Report as the shared stage). **Exercise:** participants actually facilitate the portfolio sync; executive backbrief.

Artifact: operating concept "portfolio sync" (agenda, rules, escalation criteria)

Module 6 · Months 11–12 AAR & capstone exercise

Learn AAR facilitation (4-question format, rank-free, incl. the **anti-resulting rule**: assessing decision and outcome quality separately); set up a central lessons-learned register. **Capstone:** a two-day **staff exercise** – a fictitious but realistic large scenario (e.g. "acquisition of a company, integration into the solution within 9 months") is planned end to end: situation analysis → options → wargame → decision before the real executive board → AAR.

Touchstone: does the team pass the exercise in front of real decision-makers?

D.3 Quick start – if 12 months is too big

1. **Immediately (week 1):** download the Red Team Handbook; give the next big portfolio decision a 60-minute premortem.
2. **Quarter 1:** establish the portfolio sync as an "O&I light": fortnightly, 45 minutes, the consolidated Situation Report as the only slide, the chief moderates, bad news gets praised.
3. **Quarter 2:** introduce an AAR after every PI/quarter (4 questions, rank-free, 90 minutes, documented).
4. **Quarter 3:** introduce the planning standard: no board submission for major undertakings without ≥2 real options + documented assumptions + an intent paragraph.
5. **Quarter 4:** decide on the full curriculum (D.2) based on the experience gained.

Connection to our software (memo)

- The aggregated **solution report is our O&I artifact** – traffic-light/trend logic, data-quality flags and executive summary pay directly into "shared consciousness".
- **Simulate module = quantitative wargaming:** compute disruption scenarios as parameter variants and place them side by side in the report.

- Possible later features: an **assumptions register** per epic (with expiry date/review trigger), **decision-point tracking** (early indicator + threshold + agreed response), an **AAR archive** linked to PIs/epics.

Sources

Book (new core source): McChrystal, Collins, Silverman, Fussell: *Team of Teams. New Rules of Engagement for a Complex World*. Portfolio/Penguin, 2015 (German edition: Vahlen). Concept overviews: McChrystal Group ("Shared Consciousness", "Empowered Execution"), The Field Grade Leader, Readinggraphics.

Staff/planner training: Wikipedia "Führungsakademie der Bundeswehr" and "Offizier im Generalstabsdienst" (LGAN: 2 years, ~18% selection) · Army University / Small Wars Journal on SAMS (11-month planner education, OPT assignment) · Wikipedia "Military Decision Making Process" and CALL/GlobalSecurity on COA wargaming · US Army: *Red Team Handbook* v9 / *Applied Critical Thinking Handbook* v8.1 (free PDF) · Forbes (Hoffman) on the 2021 UFMCS defunding.

Civilian transfer / programs: McChrystal Group (red-teaming workshops) · Bryce G. Hoffman: *Red Teaming* (2017), Red Team Thinking LLC · Darling/Parry/Moore: "Learning in the Thick of It", *Harvard Business Review*, July–Aug 2005 · already on the reading list: Marquet, Dörner, Vester, Meadows, Wardley, Leopold, Vacanti.

Judgment hygiene (Part A2.5, new in v4.0): Duke, Annie: *Thinking in Bets. Making Smarter Decisions When You Don't Have All the Facts*. Portfolio, 2018 — short appraisal in Lagebild-Literaturliste.md (supplement v5); picked up in the AI memorandum v1.1 (A2.6, M4 as a Ulysses contract).

Architecture thought leaders (Part C.4, new in v3.0): Hohmann, Luke: *Innovation Games* (2006) · *Beyond Software Architecture* (2003) · Tanner/Hohmann: *Software Profit Streams™* (2023). — Hohpe, Gregor: *The Software Architect Elevator* (2020) · Hohpe/Woolf: *Enterprise Integration Patterns* (2003) · architectelevators.com. — Fowler, Martin: *Patterns of Enterprise Application Architecture* (2002) · *Refactoring*, 2nd ed. (2018) · martinfowler.com. — Martin, Robert C.: *Clean Code* (2008) · *Clean Architecture* (2017). — Full source profiles and the EA transfer: Enterprise-Architektur-und-Solution-Lagebild_v2.0.md, Part A2; short appraisals in Lagebild-Literaturliste.md (supplement v3).

Version history

Version	Date	Changes
1.0	2026-08-03	Initial edition: research briefing (Team of Teams, staff/planner training, transfer to IT portfolios), 4 figures, 12-month curriculum proposal, DE + EN.
2.0	2026-08-03	Deepened edition: Part A2 with four additional core sources (Clausewitz, Marquet, Pink, Hubbard); German general-staff training deepened (BLS/SFL/LGAI), UK (JSCSC/ACSC) and Austria (National Defence Academy) added; commonalities box; civilian training market extended (calibration, superforecasting, Kepner-Tregoe, Cynefin); calibration training added to module 3; reading-list supplement v2.
2.1	2026-08-07	Typesetting/layout care per the series standard: page-break/reading-flow rules (headings stay with their following text, paragraph control, no near-empty pages – Part C no longer forces a page break). Content unchanged from 2.0.
3.0	2026-08-07	Extension: new section C.4 "Four civilian corroborating witnesses from architecture practice" – Hohmann (Innovation Games/Buy a Feature/participatory budgeting as decision rituals), Hohpe (Architect Elevator as vertical liaison; "first derivative" → trends into the picture), Fowler (pattern vocabulary as civilian proof of success factor 4; design stamina), Robert C. Martin (katas/practice culture; discipline as prerequisite for speed). Role mapping C.1, program table D.1 and curriculum module 2 extended accordingly; cross-references to the EA memorandum v2.0 (Part A2).
4.0	2026-08-13	Addition: new section A2.5 (Annie Duke, <i>Thinking in Bets</i>) – deciding as betting, resulting (decision ≠ outcome quality), "Wanna bet?" calibration, truthseeking groups, mental time travel/Ulysses contracts. Worked in: the anti-resulting rule in the AAR (B.3) and competency field 5, forecast literacy in competency field 1, the betting ritual + Ulysses contracts in module 3, the anti-resulting rule in module 6; sources section + reading-list supplement v5. Cross-reference: AI memorandum v1.1.

Source briefing "Team of Teams & Staff Training" · version 4.0 · Solution/Portfolio Management · as of 2026-08-13 · curated by Robert Seebauer, produced with Claude Fable 5 (Anthropic) · German markdown source: Quellen/Team-of-Teams-und-Stabsausbildung_v4.0.md